Project Planning & Management

Project: LinkedIn Job Market Analysis — What Drives Demand and Pay for Data Roles
Track: Junior Data Analysis — DEPI (Digital Egypt Pioneers Initiative) **Team:** [Name 1], [Name 2], [Name 3] *(replace with your team)* **Version / Date:** 1.0 — 15 Oct 2026

1. Project Proposal

1.1 Overview

This project analyses a snapshot of **123,849 LinkedIn job postings** (24 Mar – 20 Apr 2024) together with company, skill, industry, benefit and salary tables. The output is an interactive **Power BI dashboard** supported by a documented data model and DAX measures that explain what the job market asks for and what it pays.

1.2 Problem Statement

Job seekers and junior analysts struggle to see, in one place, which roles are in demand, which skills and industries pay more, and how experience level, work type and remote options change pay and competition. Raw posting data is large (about 500 MB), messy and spread across 10 linked tables, so it is hard to use without a model and a dashboard.

1.3 Objectives

1. Clean and integrate the 11 source files into one star-schema data model.
2. Measure demand by job title, industry, company, location, work type and experience level.
3. Identify the skills and benefits most requested and their relationship with salary.
4. Measure competition (views, applications, apply rate) per job category.
5. Deliver a 6-page Power BI dashboard with at least 7 documented, evidence-based insights.
6. Provide a dedicated deep-dive on Data / Business Analyst roles (408 titles contain "Data Analyst"; 4,920 contain "Analyst").

1.4 Scope

In scope: data cleaning, data modelling, DAX measures, 6 dashboard pages, insights report, testing, GitHub documentation. **Out of scope:** live data collection or scraping, predictive/ML salary models, text mining of full job descriptions (optional extension), time-trend analysis over years (the data covers about one month only).

1.5 Dataset Summary

Table	Rows	Role
postings	123,849	Fact table (jobs)
salaries	40,785	Pay details
job_skills	213,768	Job–skill link
job_industries	164,808	Job–industry link
benefits	67,943	Job benefits
companies	24,473	Company dimension
company_industries / specialities / employee_counts	24,375 / 169,387 / 35,787	Company details
industries / skills (mappings)	422 / 35	Lookup tables

1.6 Deliverables

- Project documentation (planning, literature review, requirements, analysis & design)
- Cleaned dataset and Power Query / Python cleaning scripts
- Power BI file (.pbix) with data model and DAX measures
- 6-page dashboard and insights summary
- Final presentation, test report and GitHub repository

2. Project Plan

2.1 Timeline (Gantt)

Phase	Start	End	15 Oct	Oct W4	Nov W1	Nov W2	Nov W3	Nov W4	Dec W1
Planning & Management	15 Oct	16 Oct	■						
Literature Review	15 Oct	16 Oct	■						
Requirements Gathering	15 Oct	16 Oct	■						
System Analysis & Design	17 Oct	6 Nov		■	■				
Data Cleaning & Modelling	7 Nov	16 Nov				■	■		

Phase	Start	End	15 Oct	Oct W4	Nov W1	Nov W2	Nov W3	Nov W4	Dec W1
DAX & Dashboard Development	12 Nov	30 Nov				■■	■■	■■	
Testing & Reports	25 Nov	3 Dec						■■	■■
Final Presentation	4 Dec	4 Dec							■■

2.2 Milestones

#	Milestone	Date	Evidence
M1	Planning, literature review and requirements submitted	16 Oct 2026	Documents on GitHub
M2	System analysis & design (ERD, data model, dashboard wireframes)	6 Nov 2026	Design document
M3	Clean data and star-schema model loaded in Power BI	16 Nov 2026	.pbix + cleaning scripts
M4	All DAX measures and 6 pages built	30 Nov 2026	Dashboard
M5	Testing complete, final presentation delivered	4 Dec 2026	Test report + slides

2.3 Resource Allocation

Resource	Use
Power BI Desktop	Modelling, DAX, dashboard
Python (pandas) / Power Query	Cleaning large CSV files
GitHub	Version control and document submission
Excel	Quick validation of results
Hardware	Machine with at least 16 GB RAM recommended for the 500 MB postings file
Effort	About 6–8 hours per team member per week

3. Task Assignment & Roles

Role	Responsibilities	Assigned to
Project Manager	Plan, deadlines, GitHub submissions, meetings	[Name 1]
Data Engineer	Data cleaning, outlier handling, Power Query / Python scripts	[Name 2]
Data Modeller & DAX Developer	Star schema, relationships, DAX measures	[Name 3]
Dashboard Designer	Layout, visuals, theme, interactivity	[Name 1/2/3]
QA & Documentation	Testing, insights report, final presentation	[Name 1/2/3]

For a smaller team, one member can take more than one role; the Project Manager always reviews each deliverable before submission.

4. Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
R1	Large file (about 500 MB) slows or crashes Power BI	High	High	Load only needed columns, drop description text, develop on a sample then refresh with full data
R2	Salary available for only about 29% of postings, with extreme outliers	High	High	Report salary analysis on the salary subset only, state coverage on every salary visual, filter unrealistic values
R3	Many missing values (remote flag, applies)	High	Medium	Document missing rates, avoid conclusions on columns with very low coverage
R4	Data is a one-month snapshot, mostly US-centric	Medium	Medium	State the limitation, avoid trend or Egypt-specific claims

#	Risk	Likelihood	Impact	Mitigation
R5	Many-to-many links (skills, industries) cause double counting	Medium	High	Use bridge tables, DISTINCTCOUNT on job_id, test totals
R6	Missed deadlines	Medium	High	Weekly check-in, milestones in this plan, buffer before each deadline
R7	Team member unavailable	Low	Medium	Shared GitHub repo, two people familiar with each task
R8	Data licensing / privacy	Low	Medium	Use public Kaggle data only, no personal data, cite the source

5. Key Performance Indicators (KPIs)

KPI	Target
Milestones delivered on schedule	100%
Source tables integrated into the model	10 of 10
Duplicate job_id in the fact table	0
DAX measures built and documented	25 or more
Dashboard pages	6
Dashboard page load / filter response time	Under 5 seconds
Documented, evidence-based insights	7 or more
Reconciliation: dashboard job count equals source count	100% match
Test cases passed	95% or higher
Documents uploaded to GitHub on time	100%